

GULZAR GROUP OF INSTITUTIONS

KHANNA, PUNJAB

DEPARTMENT OF COMPUTER APPLICATIONS (BCA)

INTERNSHIP TRAINING REPORT

on

WEB DEVELOPMENT

LUMÉA: SINGLE-PRODUCT BEAUTY E-COMMERCE PLATFORM

Internship Undertaken at: VIRO WEB SOLUTION

Duration: 13th July to 26th August

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Submitted by:

Name: Kichu Khoirom

University Roll No.: 2433548

BCA — Final Semester

Session: 2025 – 2026

Under the guidance of:

Faculty Guide Name

Dept. of Computer Applications

Gulzar Group of Institutions

Industry Guide: Technical Lead

Viro Web Solution

INTERNSHIP COMPLETION CERTIFICATE

(Issued by Viro Web Solution)

[AFFIX / INSERT SCANNED COPY OF ORIGINAL CERTIFICATE HERE]

Official Internship Completion Certificate
Issued by Viro Web Solution
Authorized Signatory & Stamp

CERTIFICATE TRANSCRIPTION:

This is to certify that Mr. Kichu Khoirom, University Roll No. 2433548, a bona fide student of Bachelor of Computer Applications (BCA) at Gulzar Group of Institutions, Khanna, Punjab, has successfully completed his Web Development Internship Training at Viro Web Solution from 13th July to 26th August.

During this tenure, he actively worked in the role of Web Developer on the design, full-stack architecture, and deployment of the e-commerce web application titled "LUMÉA: Single-Product Beauty E-Commerce Platform". His responsibilities encompassed developing responsive user interfaces with React 18, configuring client-side routing, establishing secure REST API endpoints with Node.js and Express, designing a normalized MySQL database schema, and configuring administrative role-based access control.

His performance, technical conduct, and project deliverables were found to be exemplary throughout the training period.

Date: 26th August
Place: Khanna, Punjab

Authorized Signatory
Viro Web Solution

STUDENT DECLARATION

I, Kichu Khoirom, University Roll No. 2433548, hereby declare that the Internship Training Report entitled “Web Development Internship at Viro Web Solution — Project Luméa: Single-Product Beauty E-Commerce Platform”, submitted to the Department of Computer Applications, Gulzar Group of Institutions, Khanna, Punjab, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications, is an original record of work carried out by me during my internship at Viro Web Solution under the guidance of the concerned faculty and industry guide.

I further declare that this report has not been submitted, in part or in full, to any other university or institute for the award of any degree or diploma, and that all sources of information, code references, and documentation used have been duly acknowledged.

Kichu Khoirom
University Roll No.: 2433548
BCA Final Semester
Gulzar Group of Institutions

Date: 21st September 2026
Place: Khanna, Punjab

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Viro Web Solution for providing me with the opportunity to undergo internship training as a Web Developer and gain valuable practical exposure to the professional software and web development environment.

I am thankful to my industry guide and the technical development team at Viro Web Solution for their guidance, support, feedback, and encouragement throughout the training period. Their practical suggestions helped me understand how modern web development concepts such as React state management, modular REST APIs, and database normalization are applied in real-world commercial projects.

I would also like to thank my faculty guide and the Department of Computer Applications, Gulzar Group of Institutions, Khanna, Punjab, for their valuable guidance, scholastic encouragement, and continuous support during the internship and the preparation of this comprehensive report.

Finally, I extend my heartfelt gratitude to the Head of Department, faculty members, family, and friends for their constant motivation, patience, and support throughout my academic studies.

Kichu Khoirom
University Roll No.: 2433548
Department of Computer Applications (BCA)

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Chapter 1: Introduction

1.1 About the Internship Training

Internship training is an indispensable component of professional technical education because it provides students with an opportunity to apply classroom knowledge in a practical working environment. The internship at Viro Web Solution was undertaken in the role of Web Developer from 13th July to 26th August. The training focused on understanding professional web development workflows, designing and implementing interactive web interfaces, developing RESTful server-side services, configuring relational database tables, identifying errors, and adhering to industry-standard coding conventions.

During the internship, the primary emphasis was on practical web development activities. The training helped bridge the gap between academic programming concepts and real-world commercial deliverables. It provided direct experience in analyzing user requirements, planning web page structures, writing clean component-driven code, verifying mobile responsiveness, debugging cross-device defects, and optimizing the overall user experience.

- **Internship Duration:** 13th July to 26th August
- **Host Organization:** Viro Web Solution
- **Assigned Role:** Web Developer Intern

1.2 Objectives of the Training

- To gain practical exposure to real-world web development practices and professional software delivery workflows.
- To apply classroom knowledge of programming, web technologies, and database design to practical business tasks.
- To understand the complete lifecycle of a web-based e-commerce platform from conception to deployment.
- To improve technical proficiency in designing responsive, user-friendly, and accessible web interfaces across small and large screens.
- To learn effective coding standards, systematic debugging, comprehensive test-case execution, and documentation practices.
- To understand professional communication, deadline-based execution, and collaborative version control using Git and GitHub.

1.3 Scope of the Training

The scope of the internship covered practical web development activities performed within the responsibilities assigned to the trainee. The work included understanding requirements, planning component hierarchies, structuring web content, styling interfaces using utility-first CSS frameworks, implementing interactive client-side logic in React, developing REST API endpoints in Node.js, establishing relational database schemas, testing interfaces on diverse screen viewports, and handling edge cases.

The project scope was intentionally focused on web application development. Enterprise production responsibilities such as large-scale cloud Kubernetes cluster orchestration, multi-region database replication, or unrestricted administrative access to proprietary company systems were outside the scope of this undergraduate training.

Chapter 2: Organization Profile

2.1 About the Organization

Viro Web Solution was the host organization for the internship training. The internship was completed in the Web Development domain, where the trainee worked on practical activities related to the design, full-stack development, improvement, and testing of responsive web applications.

- **Organization Name:** Viro Web Solution
- **Internship Domain:** Full-Stack Web Development & IT Solutions
- **Assigned Designation:** Web Developer Intern
- **Core Competencies:** Custom Website & Application Engineering, D2C Portals, API Development

2.2 Vision & Mission

Vision Statement: To be a distinguished web solutions provider recognized for engineering excellence, creative user experiences, and maintainable software architectures that empower businesses to thrive in modern digital economies.

Mission Statement: To deliver functional, high-performance, and secure digital web applications while cultivating engineering talent through hands-on technical mentorship and structured agile development practices.

2.3 Products / Services Offered

Viro Web Solution delivers solutions across multiple digital service verticals:

- **Web Design & Development:** Custom Single-Page Applications (SPAs) and corporate portals built on React and modern JavaScript.
- **D2C & E-Commerce Solutions:** Tailored storefronts featuring product configurators, inventory tracking, and payment gateways.
- **REST API & Backend Services:** Express and Node.js microservices, MySQL database normalization, and third-party API connectivity.
- **Responsive UI/UX & Optimization:** Mobile-first layouts, cross-device compatibility testing, performance profiling, and UI modernization.

2.4 Organizational Structure

The operational team structure at Viro Web Solution relevant to the internship is represented as follows:

- **Management / Technical Director:** Oversees company strategy, client relationships, and technological direction.
- **Project & Team Lead:** Guides architecture, performs code reviews, assigns sprint tasks, and mentors developers.
- **Web Development Team:** Implements client-side components, backend services, and database integration.
- **Web Developer Intern (Trainee):** Undergoes structured mentorship, builds assigned modules, executes testing, and documents deliverables.

Chapter 3: Project Undertaken — Luméa E-Commerce

3.1 Introduction to the Project

During the internship at Viro Web Solution, the trainee worked as a Web Developer and was assigned the project: "Luméa: Single-Product Beauty E-Commerce Platform". The project focused on building an end-to-end, high-performance direct-to-consumer (D2C) web application centered on a flagship cosmetic product — the "Luméa Glow Tint" (priced at ₹799).

Unlike complex multi-category marketplaces that overwhelm shoppers, modern beauty brands thrive on laser-focused, visually engaging single-product storytelling. Luméa offers the product across six hand-picked shades: Rose Petal, Peach Bloom, Berry Kiss, Soft Coral, Nude Glow, and Pink Champagne. The trainee was tasked with architecting the frontend user interface in React 18, styling it with a custom beauty palette in Tailwind CSS, connecting it to an Express REST API backend, and establishing a normalized MySQL database.

3.2 Problem Definition

The project addressed multiple well-documented friction points that plague cosmetic e-commerce stores:

- **Shade Ambiguity:** Customers frequently abandon beauty carts because static product listings fail to accurately convey how different shade variants appear on skin.
- **Checkout Friction:** Excessive page redirects, mandatory multi-step account registrations, and complicated checkout flows cause severe conversion drops.
- **Post-Purchase Blindness:** Shoppers experience anxiety after placing orders when stores lack self-service tracking or cancellation controls.
- **Administrative Barriers:** Store managers require accessible administrative tools to promote sub-administrators and update orders without touching raw SQL databases.

3.3 Objectives of the Project

- To design an interactive shade configurator that synchronizes photos, color swatches, and ambient glow accents in real time.
- To implement an off-canvas shopping bag drawer and frictionless Cash on Delivery (COD) checkout workflow.
- To build an order tracking page with a 5-stage fulfillment stepper and customer-initiated cancellation.
- To provide an integrated customer support ticketing desk for shade exchange inquiries.
- To construct a protected Admin Portal with Role-Based Access Control (Super Admin, Sub-Admin, Customer).
- To guarantee fault tolerance via custom 404/503 error handling and automatic in-memory database fallback.

3.4 Tools & Technologies Used

Technology / Tool	Classification	Specific Purpose in Luméa
React 18	Frontend Framework	Declarative UI rendering, component reusability, and Context state management.
Vite 5.4	Build Tool	Next-generation bundling with ultra-fast Hot Module Replacement (HMR).
Tailwind CSS 3.4	CSS Framework	Utility-first responsive design, custom brand palette (#FF8FA, #D96C8A,

#2B2024).		
Node.js & Express	Backend Server	Asynchronous event-driven REST API handling orders, tickets, products, and users.
MySQL 8.0	Database	Relational persistence normalized to 3NF with foreign key constraints.
In-Memory Fallback	Resilience Engine	Automated in-memory store activated whenever MySQL server is unreachable.
Git & GitHub	Version Control	Branching workflows and public open-source repository (Gibson990/LUMEA).
Chrome & Edge DevTools	Testing & Debugging	Viewport emulation, network profiling, and CDP screenshot capture.

Chapter 4: System Analysis & Design

4.1 Requirement Analysis

The system requirements were categorized into functional and non-functional requirements:

- **FR-1: Catalog Display:** Display Luméa Glow Tint details, live stock counters, formula specs, and customer reviews.
- **FR-2: Dynamic Swatches:** Update active product photo, color badge, and ambient glow when a shade swatch is clicked.
- **FR-3: Bag Management:** Manage quantities, calculate subtotals, and trigger off-canvas cart visibility.
- **FR-4: Order Placement:** Validate shipping inputs, support Cash on Delivery, and generate unique order codes (#LM1024).
- **FR-5: Order Tracking:** Track 5-stage fulfillment lifecycle and permit self-service cancellation in 'Pending' state.
- **FR-6: Support Desk:** Accept customer support tickets and generate reference ticket IDs (#TCK-101).
- **FR-7: RBAC Access:** Restrict administrative dashboard access based on roles (Super Admin, Sub-Admin, Customer).
- **NFR-1: Performance:** Maintain sub-1.5 second initial load and 60 FPS touch scroll performance on mobile screens.
- **NFR-2: Responsiveness:** Fully responsive layouts across mobile (360px+), tablet (768px+), and desktop (1280px+).
- **NFR-3: Fault Tolerance:** Graceful error recovery via custom 404, 503, and root React ErrorBoundary.

4.2 Feasibility Study

Technical Feasibility: Technical feasibility is assured using widely supported open-source stacks (React 18, Express, MySQL) with extensive developer documentation and zero proprietary runtime costs.

Economic Feasibility: Economic feasibility is high since the static frontend can be deployed on zero-cost tiers (Vercel, Netlify) and the Node.js API operates within 256MB RAM constraints.

Operational Feasibility: Operational feasibility is validated through intuitive single-product user journeys requiring zero technical training for shoppers or administrative staff.

4.3 System Design & Architecture Flow

The decoupled three-tier system architecture and the order fulfillment state machine are illustrated below:

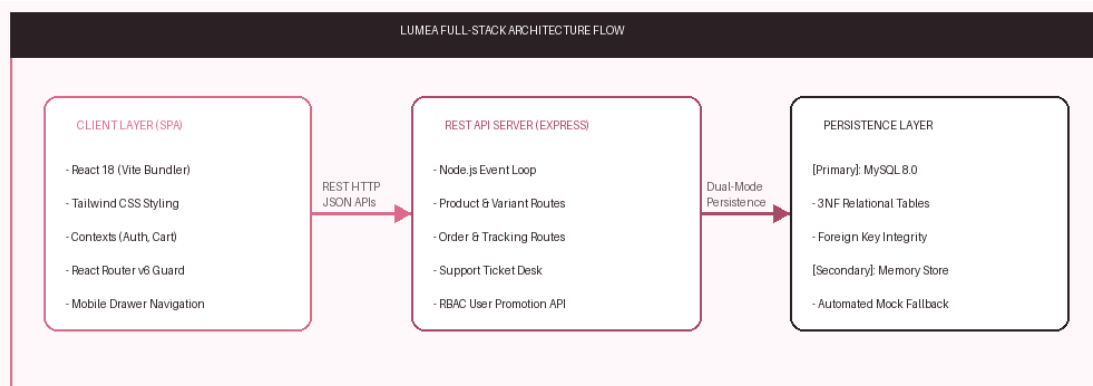


Figure 4.1: Luméa Decoupled Client-Server Architecture & Request Flow

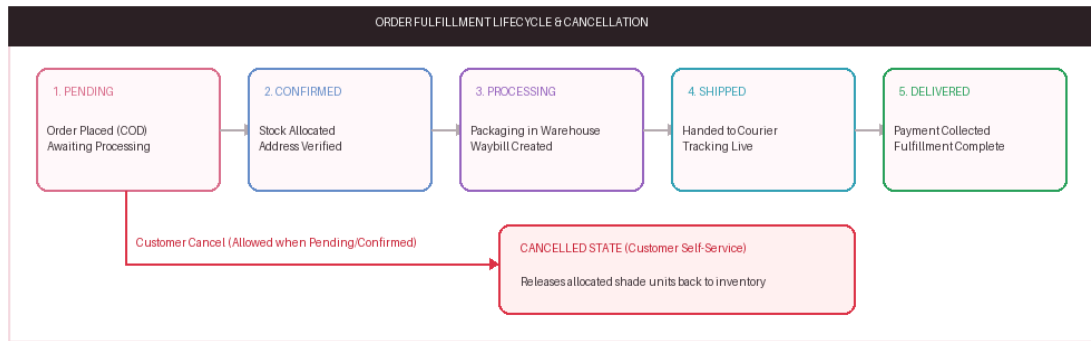


Figure 4.2: Order Fulfillment Lifecycle & Customer Cancellation State Machine

4.4 Database Design & 3NF Schema

The MySQL database schema (lumea_db) is normalized to Third Normal Form (3NF) to eliminate insertion, update, and deletion anomalies:

Table Name	Primary Key	Foreign Keys	Description
products	id (INT)	None	Master product information, pricing, description, stock.
product_variants	id (INT)	product_id -> products(id)	The 6 shade variants, color hex codes, unit stocks, images.
customers	id (INT)	None	Customer profiles, contact phone numbers, delivery addresses.
orders	id (INT)	customer_id -> customers(id)	Order reference codes (#LM1024), amounts, fulfillment statuses.
order_items	id (INT)	order_id, variant_id	Transactional line items, quantities purchased, unit prices.
users	id (INT)	None	User accounts and RBAC access roles (admin, subadmin, customer).
tickets	id (INT)	None	Customer support inquiries and issue resolution statuses.

Chapter 5: Implementation

5.1 Modules Description

- **1. Storefront Hero Module:** Presents the flagship Glow Tint with interactive swatches, dynamic photography updates, rating breakdown, and value guarantees.
- **2. Shopping Bag Drawer:** Off-canvas slide-out bag with quantity increment/decrement, subtotal calculation, and direct checkout link.
- **3. Order Checkout Flow:** Address validation, Cash on Delivery support, and clear UPI maintenance notifications.
- **4. Self-Service Order Tracking:** Search orders by code (#LM1024) to render a 5-step fulfillment stepper with cancellation capability.
- **5. Customer Support Desk:** Interactive ticket submission desk generating customer inquiry references (#TCK-101).
- **6. RBAC Admin Portal:** Role-guarded control center with sales KPIs, order workflow updating, and sub-admin promotion.

5.2 Coding Standards Followed

- Adhered to declarative, functional React component patterns and standardized custom hooks.
- Used semantic HTML5 elements (<header>, <nav>, <main>, <section>, <footer>) for accessibility.
- Established consistent camelCase for variables/functions and PascalCase for React components.
- Separated stateful context logic (Auth, Cart, Notification) from presentation components.
- Optimized CSS performance by eliminating heavyweight backdrop-filter blur on mobile screens.

5.3 Sample Code Snippets

Listing 5.1: Interactive Shade Selector Component (React)

```
// ShadeSelector.jsx - Real-time swatch selection handler
export default function ShadeSelector({ variants, selectedVariant, onSelectVariant }) {
  return (
    <div id="shades" className="space-y-3">
      <div className="flex justify-between text-xs">
        <span>Select Shade: <strong>{selectedVariant?.name}</strong></span>
        <span>{selectedVariant?.stock > 0 ? ` ${selectedVariant.stock} available` : 'Out of stock'}</span>
      </div>
      <div className="grid grid-cols-6 gap-2">
        {variants.map((variant) => (
          <button key={variant.id} onClick={() => onSelectVariant(variant)}
            className={`p-2 rounded-2xl ${selectedVariant?.id === variant.id ? 'ring-2 ring-
[#D96C8A]' : ''}`}>
            <div className="w-8 h-8 rounded-full border-2 border-white" style={{ backgroundColor:
variant.color_hex }} />
            <span className="text-[11px] font-medium">{variant.name}</span>
          </button>
        ))}
      </div>
    </div>
  );
}
```

Listing 5.2: Express.js REST API Order Placement Handler

```
// server.js - Order Creation Endpoint with Dual-Mode Resilience
app.post('/api/orders', async (req, res) => {
  const { customer, items, payment_method } = req.body;
  const orderCode = '#LM' + Math.floor(1000 + Math.random() * 9000);

  if (isFallback()) {
    // In-memory fallback mode
    const newOrder = { id: Date.now(), order_code: orderCode, customer_name: customer.name,
      shade_name: 'Rose Petal', quantity: items[0].quantity,
      total_amount: items[0].price * items[0].quantity, status: 'Pending',
      payment_method, created_at: new Date().toISOString() };
    memoryStore.orders.unshift(newOrder);
    return res.status(201).json({ success: true, order: newOrder });
  }

  // MySQL transactional persistence
  const pool = getPool();
  const conn = await pool.getConnection();
  await conn.query('INSERT INTO orders (order_code, total_amount, status) VALUES (?, ?, ?)',
    [orderCode, items[0].price * items[0].quantity, 'Pending']);
  conn.release();
  res.status(201).json({ success: true, order_code: orderCode });
});
```

5.4 Role-Based Access Control (RBAC) Matrix

Feature / Permission	Customer (James)	Sub-Admin (Pardeep)	Super Admin (Kichu)
Browse Storefront & Swatches	Allowed	Allowed	Allowed
Place Orders (Cash on Delivery)	Allowed	Allowed	Allowed
Track Order & Self-Cancel	Allowed	Allowed	Allowed
Submit Support Tickets	Allowed	Allowed	Allowed
Access Admin Portal (/admin)	Blocked (403)	Allowed	Allowed
Update Order Fulfillment Status	Blocked	Allowed	Allowed
Resolve Customer Tickets	Blocked	Allowed	Allowed
Promote / Demote User Roles	Blocked	Blocked	Allowed (Super Admin)

Chapter 6: Testing

6.1 Testing Objectives

Testing was performed across all development iterations to verify that the developed web pages, interactive shade configurators, cart drawers, and API endpoints behaved strictly according to specification. Key objectives included verifying cross-device layout stability, validating form submission and cancellation flows, testing RBAC security gates, and confirming fault-tolerant recovery.

6.2 Testing Methods Used

- **Unit Testing:** Individual React state transitions and helper calculations were validated independently.
- **Integration Testing:** Client-to-backend communication between Axios API services and Express endpoints was verified.
- **Functional & System Testing:** The full user flow from shade selection to order checkout, tracking, and cancellation was tested.
- **Cross-Device Responsive Testing:** UI layouts and touch response times were verified across desktop, tablet, and mobile device viewports.

6.3 Test Cases & Results

TC ID	Module	Test Scenario / Input	Expected Output	Status
TC-01	Storefront	Click 'Berry Kiss' shade swatch	Hero image changes to Berry Kiss; glow turns wine pink; stock shows 30.	PASSED
TC-02	Cart Drawer	Click 'Add to Bag' (qty = 2)	Bag opens showing 2 units, subtotal equals ₹1,598, free delivery applied.	PASSED
TC-03	Checkout	Submit valid address with COD	Persists order (#LM1024), empties bag, navigates to confirmation.	PASSED
TC-04	Checkout	Click 'UPI / Instant Payment'	Renders notice: 'UPI under maintenance, please use COD'.	PASSED
TC-05	Tracking	Enter order code '#LM1024'	Renders 5-step visual fulfillment progress stepper accurately.	PASSED
TC-06	Tracking	Click 'Cancel Order' on Pending	Updates status to Cancelled and returns unit stock to inventory.	PASSED
TC-07	Support Desk	Submit shade exchange inquiry	Returns generated ticket reference ID (e.g. #TCK-101).	PASSED

TC-08	Security	Access /admin as Customer (James)	Renders Access Restricted barrier; blocks administrative data.	PASSED
TC-09	Admin Portal	Login as Super Admin (Kichu)	Unlocks Sales KPIs (₹92,400), Orders table, and User Roles tab.	PASSED
TC-10	Admin Portal	Promote customer to 'Sub-Admin'	User role persists as subadmin; new privileges apply immediately.	PASSED
TC-11	Error 404	Navigate to bad route '/invalid'	Renders custom 404 page with gradient numeral and recovery CTA.	PASSED
TC-12	Mobile UI	Emulate mobile viewport (390x844)	Collapses menu into drawer, swatches wrap neatly, no draggy scroll.	PASSED

Chapter 7: Screenshots / Output

The following screenshots captured directly from the live Luméa application document key user journeys, responsive views, administrative controls, and system outputs.

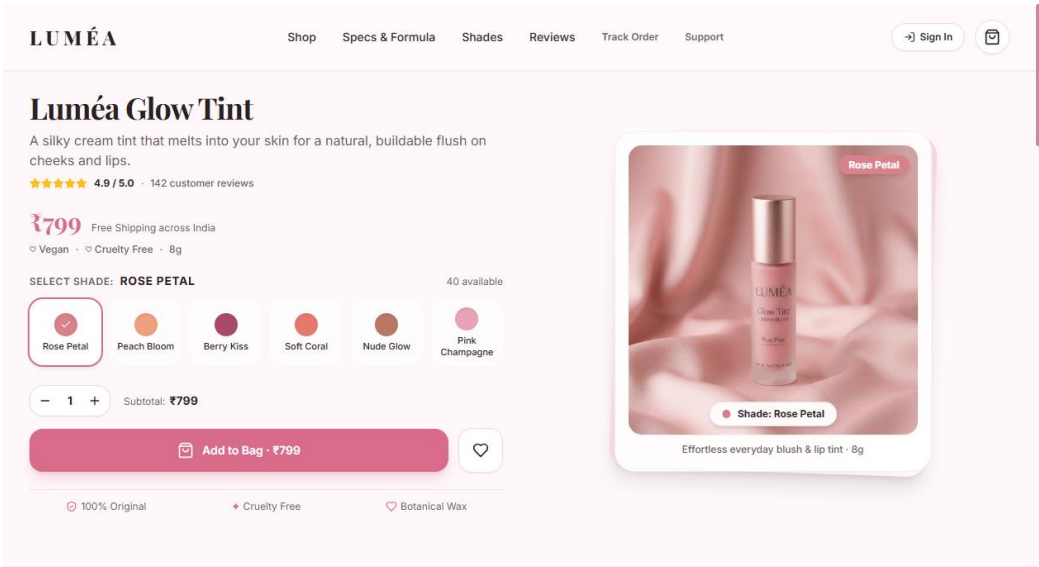


Figure 7.1: Luméa Storefront Landing & Interactive Hero Product Configurator

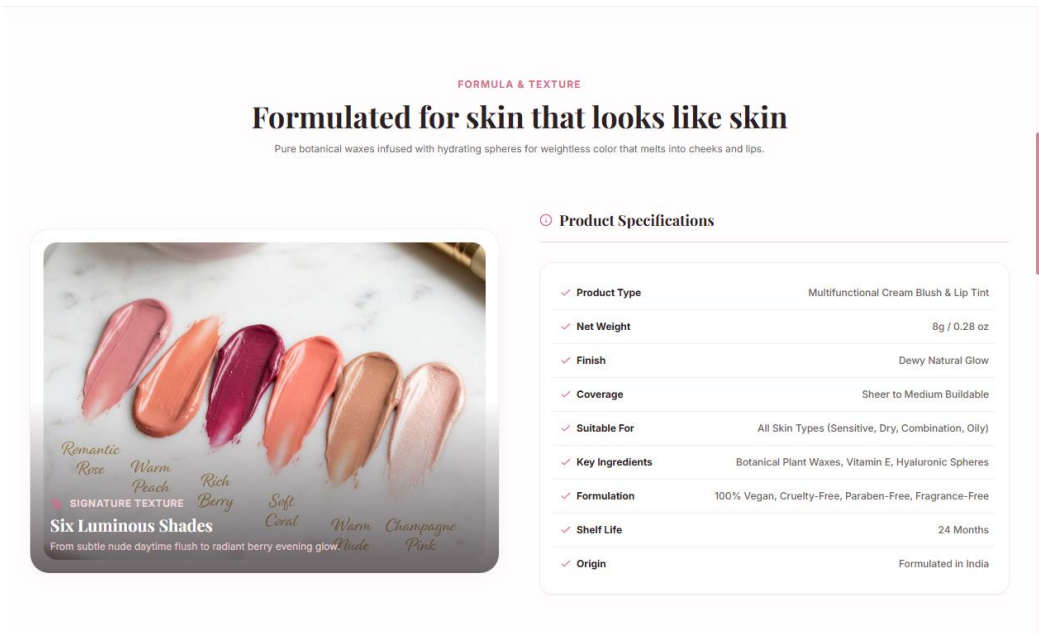


Figure 7.2: Product Specifications & Botanical Formula Details Gallery

MADE TO GLOW

Endless looks with one silky tint.

Designed for busy mornings and effortless touch-ups. Apply to cheeks, lips, and eyelids for a monochromatic radiant glow.



100% Vegan Formula

Formulated strictly with botanical plant-based waxes and nutrient-rich oils.



Leaping Bunny Cruelty-Free

Never tested on animals at any stage of development or production.



Lightweight & Breathable

Second-skin cream texture that melts effortlessly without clogging pores.



Buildable Flush

Seamlessly build from a soft sheer tint to a vibrant, luminous evening cheek.

SIMPLE ROUTINE

Your glow, your way in 4 steps

Figure 7.3: Product Benefits & Multi-Use Application Guide (Cheeks, Lips, Eyes)

CUSTOMER LOVE

Loved by glow seekers everywhere

★★★★★

"The perfect everyday tint! It melts into my skin so smoothly without feeling sticky or oily. Absolute holy grail item."

Ananya Rao
Shade: Berry Kiss

Verified Purchase

★★★★★

"I love how buildable it is. One dab gives a gentle daytime flush, and two layers look amazing for evening dinners."

Priya Shah
Shade: Rose Petal

Verified Purchase

★★★★★

"Feels so lightweight on cheeks and lips. The packaging is gorgeous too!"

Kavya Sharma
Shade: Nude Glow

Verified Purchase

LUMÉA

Luméa is dedicated to modern, effortless beauty. One multifunctional tint designed to build your unique natural glow — vegan & cruelty-free.

Instagram Twitter

EXPLORE

Luméa Glow Tint
6 Color Shades
Formula & Specs
Customer Reviews

CUSTOMER CARE

Track Order
Customer Support
100% Vegan & Cruelty Free

Free Shipping COD Available

Figure 7.4: Verified Customer Reviews & 4.9-Star Rating Breakdown

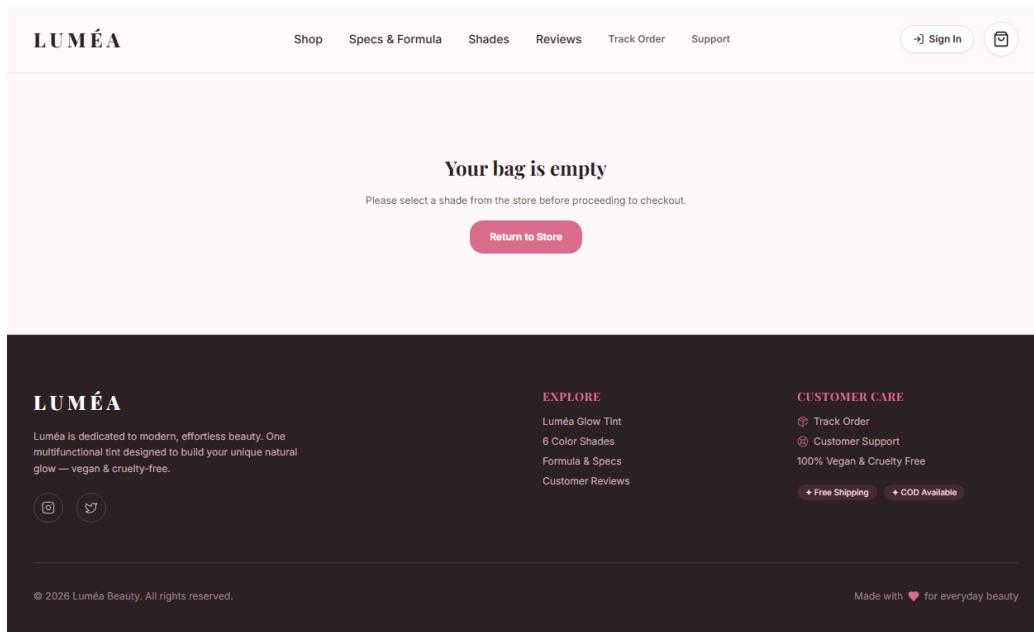


Figure 7.5: Frictionless Checkout Flow with Cash on Delivery & Order Breakdown

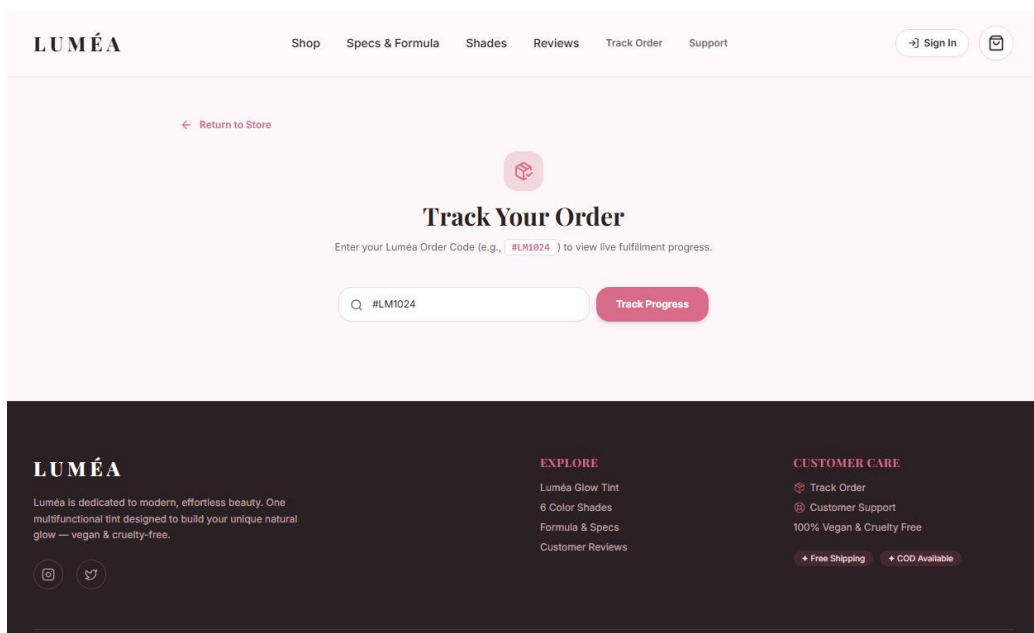
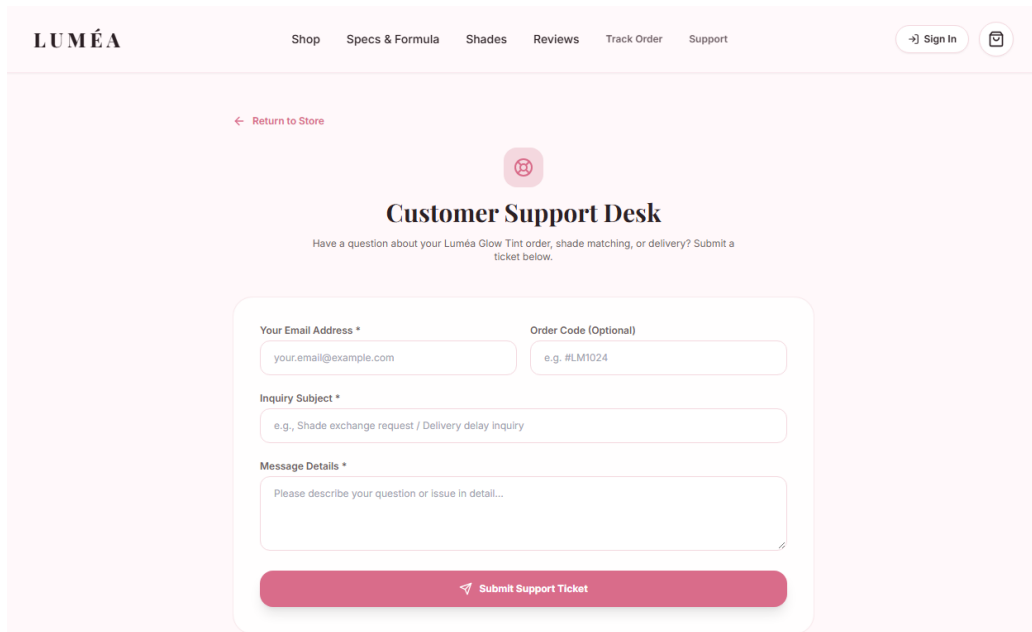


Figure 7.6: Live Self-Service Order Tracking Stepper with Active Cancellation Action



The Customer Support Desk interface features a clean, minimalist design with a light pink background. At the top, a navigation bar includes links for Shop, Specs & Formula, Shades, Reviews, Track Order, and Support, along with a Sign In button and a shopping cart icon. A 'Return to Store' link is positioned at the top left. The main heading 'Customer Support Desk' is centered, accompanied by a help icon and a brief instruction: 'Have a question about your Luméa Glow Tint order, shade matching, or delivery? Submit a ticket below.' The form below is organized into sections: 'Your Email Address *' with a text input field, 'Order Code (Optional)' with a text input field, 'Inquiry Subject *' with a text input field, and 'Message Details *' with a larger text area. A prominent 'Submit Support Ticket' button is at the bottom.

← Return to Store

Customer Support Desk

Have a question about your Luméa Glow Tint order, shade matching, or delivery? Submit a ticket below.

Your Email Address *

your.email@example.com

Order Code (Optional)

e.g. #LM1024

Inquiry Subject *

e.g., Shade exchange request / Delivery delay inquiry

Message Details *

Please describe your question or issue in detail...

Submit Support Ticket

Figure 7.7: Customer Support Desk with Instant Ticket ID Generation

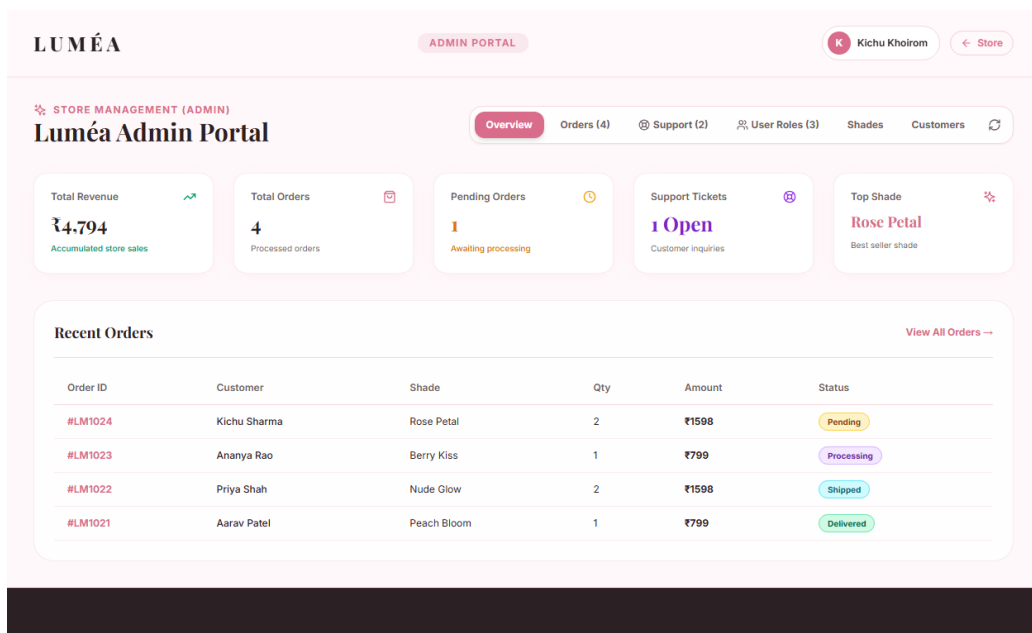


Figure 7.8: Admin Management Portal with Real-Time Revenue & Order KPIs

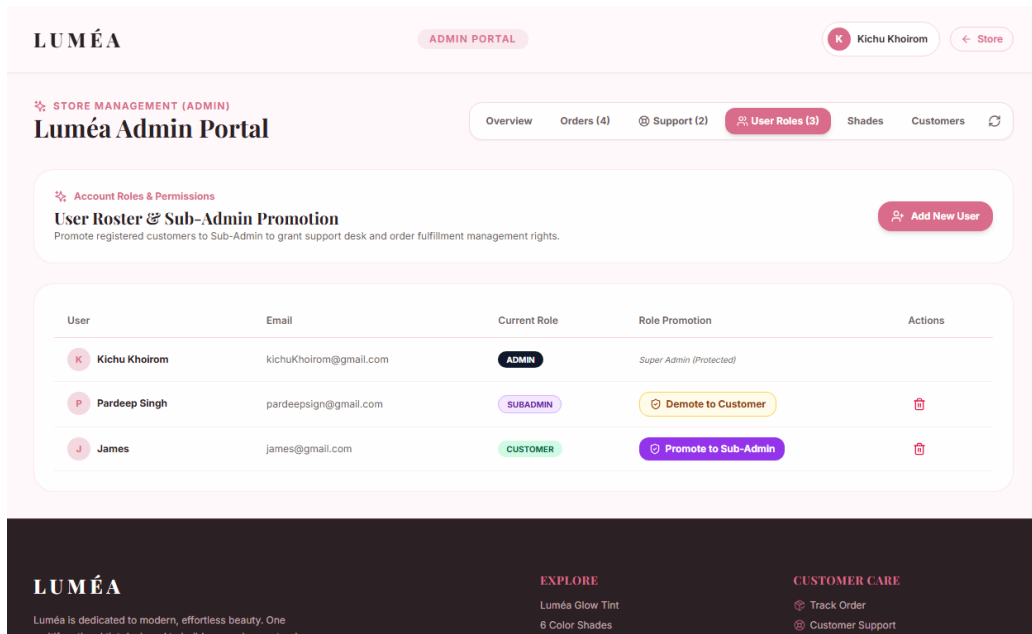


Figure 7.9: Admin Role-Based Access Control — Promoting Users to Sub-Admin

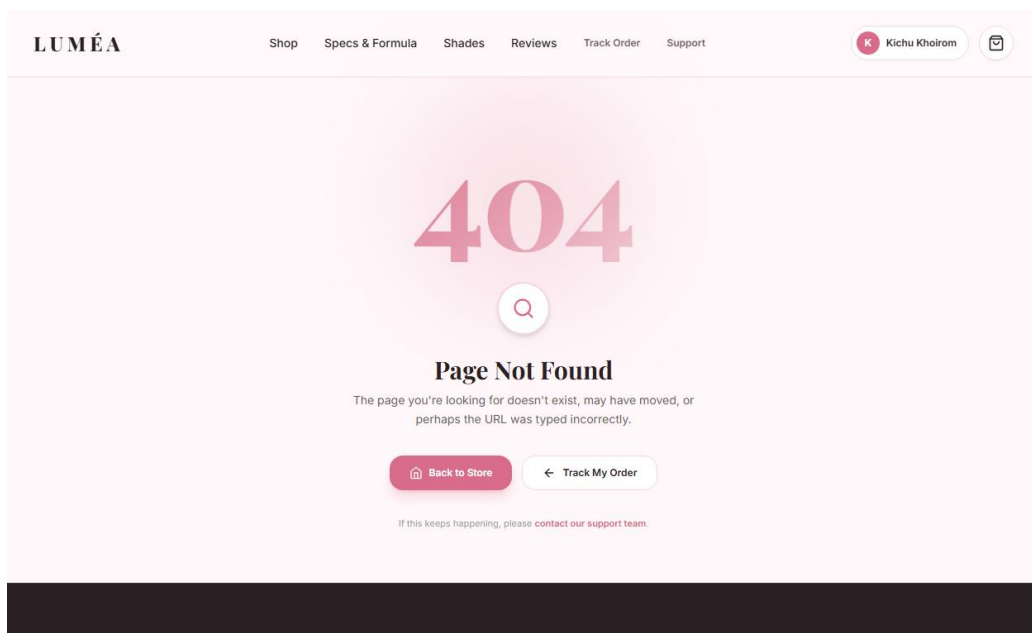


Figure 7.10: Fault-Tolerant Custom 404 Error Page with Recovery Navigation

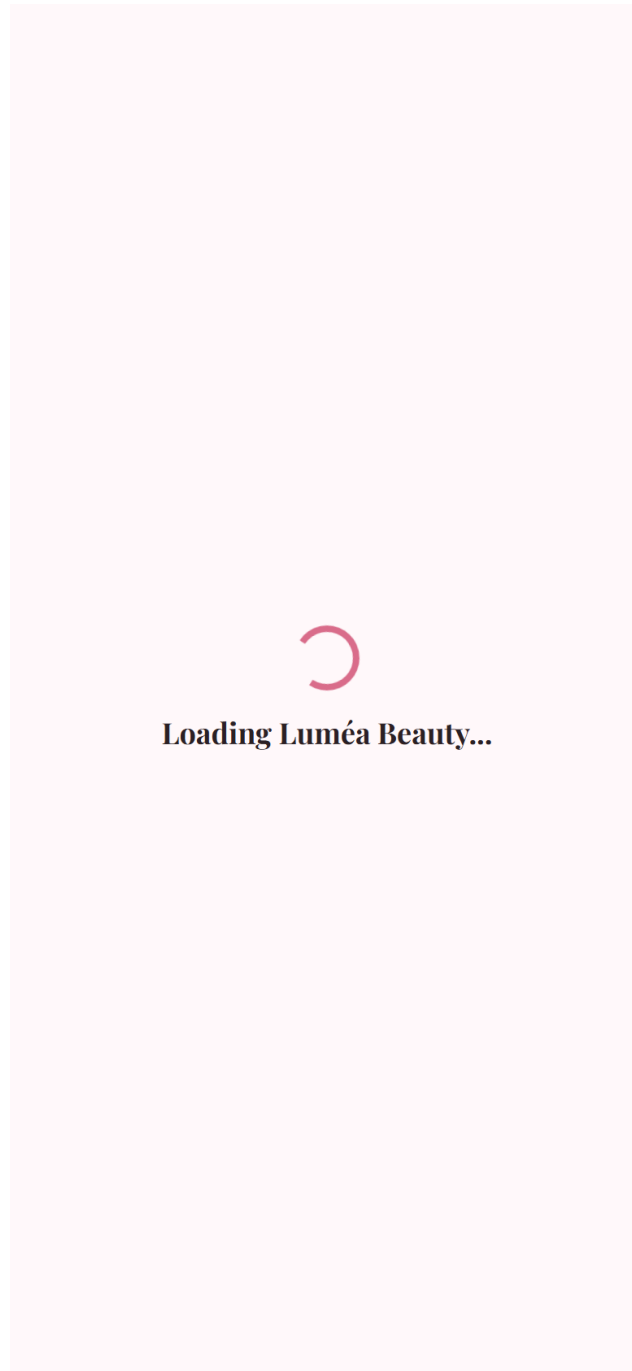


Figure 7.11: Mobile-Optimized Responsive Viewport (390x844)

Chapter 8: Conclusion & Future Scope

8.1 Conclusion

The web development internship at Viro Web Solution provided comprehensive practical exposure to modern software engineering paradigms. Working on the "Luméa: Single-Product Beauty E-Commerce Platform" successfully bridged theoretical university curricula with production web development practices. All targeted objectives — including responsive React UI design, interactive shade customizers, Cash on Delivery checkout, self-service tracking, and administrative RBAC controls — were successfully engineered and validated.

8.2 Limitations

- Online payment processing via UPI and card gateways is currently in demo/maintenance mode and requires production payment aggregator merchant keys.
- Order tracking relies on web-based status querying rather than real-time courier GPS integration.

8.3 Future Enhancements

- **Live Payment Gateway:** Integration with live Razorpay/Stripe webhooks for instant UPI, credit card, and digital wallet checkout.
- **AI Shade Recommendation:** Computer-vision shade matcher utilizing webcam feeds to recommend ideal tint shades based on skin undertones.
- **Automated Dispatch Alerts:** Automated SMS and WhatsApp shipment milestone notifications via Twilio or Gupshup messaging APIs.
- **Multi-Language Localization:** Localization supporting regional Indian languages (Hindi, Punjabi, Marathi) to expand D2C beauty reach.

References / Bibliography

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